Lecture 8. Limit Theorems

Markov and Chebyshev, weak and strong laws, central limit theorem, Chernoff bounds, Jensen

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8.1 Markov and Chebyshev inequalities; the weak law

Motivation. Markov and Chebyshev bound tails from a mean or variance alone, and give the weak law.

Proposition 8.1 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq E[X]/a$.

Proof. $a\mathbb{1}\{X \geq a\} \leq X$; take expectations. ∎

Proposition 8.2 (Chebyshev's inequality)

If X has mean μ and variance σ^2, then $P(|X - \mu| \geq k) \leq \sigma^2/k^2$ for $k > 0$.

Proof. Apply Markov's inequality to $(X - \mu)^2$ and $a = k^2$. ∎

Crude, but valid for every distribution.

Example 8.1

(a) Output with mean 50: $P(X \geq 75) \leq 2/3$; with $\mathrm{Var}(X) = 25$ also $P(40 < X < 60) \geq 3/4$.
(b) $X \sim U(0, 10)$: Chebyshev gives $P(|X - 5| > 4) \leq (25/3)/16 \approx 0.52$; truly 0.2.
(c) $\mathrm{Var}(X) = 0$ gives $P(|X - \mu| > 1/n) = 0$ for all n, so $P(X = \mu) = 1$.

Theorem 8.3 (Weak law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ, then for every $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ as $n \to \infty$.

Proof. (Finite σ^2.) $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$; Chebyshev gives $\sigma^2/(n\epsilon^2) \to 0$. ∎

8.2 The central limit theorem

Motivation. Fluctuations of averages are approximately normal, whatever the distribution.

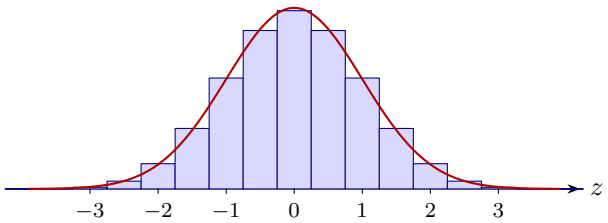


Figure. $(S_{16} - 8)/2$ *for* $S_{16} \sim \mathrm{Bin}(16, \frac{1}{2})$ *(bars, unit area) against the* $N(0, 1)$ *density.*

Theorem 8.4 (Central limit theorem)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, then for every a,

$$P\Big(\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \leq a\Big) \to \Phi(a) \qquad (n \to \infty).$$

Key tool (quoted): $M_{Z_n}(t) \to M_Z(t)$ for all t implies $F_{Z_n}(a) \to F_Z(a)$ at continuity points.

Proof sketch. Take $\mu = 0$, $\sigma = 1$, mgf M near 0, $L = \log M$: $L(0) = 0$, $L'(0) = 0$, $L''(0) = 1$. $Z_n = \sum X_i/\sqrt{n}$ has mgf $M(t/\sqrt{n})^n$, and by L'Hôpital twice ($x = 1/\sqrt{n}$)

$$nL\Big(\frac{t}{\sqrt{n}}\Big) = \frac{L(tx)}{x^2} \to \frac{t^2 L''(0)}{2} = \frac{t^2}{2}.$$

So $M_{Z_n}(t) \to e^{t^2/2}$. (In general use characteristic functions.) ■

Hence sums of many small effects are approximately normal. The CLT also holds for independent uniformly bounded X_i with $\sum \mathrm{Var}(X_i) = \infty$: $(\sum(X_i - \mu_i))/\sqrt{\sum \sigma_i^2} \approx N(0,1)$.

Example 8.2 (How many measurements?)

Averaging n measurements of variance 4, to be 95% sure of accuracy 0.5: CLT needs $0.25\sqrt{n} \geq 1.96$, $n \geq 62$; Chebyshev needs $4/(0.25n) \leq 0.05$, $n \geq 320$ (valid for any distribution).

Example 8.3

(a) 10 fair dice: mean 35, variance 350/12; with continuity correction $P(30 \leq S \leq 40) \approx 2\Phi(5.5/5.40) - 1 \approx 0.69$.
(b) 10 uniforms: mean 5, variance 10/12, $P(\sum X_i > 6) \approx 1 - \Phi(1.10) \approx 0.137$.
(c) Grading times with mean 20, SD 4 minutes: 25 exams within 450 minutes has probability $\approx \Phi(-2.5) \approx 0.006$.

8.3 The strong law of large numbers

Motivation. The strong law makes the whole sequence of averages converge almost surely, justifying frequencies.

Theorem 8.5 (Strong law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with finite mean μ, then $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$.

Proof. Assume $K = E[X_i^4] < \infty$, $\mu = 0$. In $E[S_n^4]$ terms with an index appearing once vanish, leaving $nE[X^4] + 6\binom{n}{2}E[X_i^2 X_j^2] \leq nK + 3n(n-1)K$. So $E[\sum_n S_n^4/n^4] \leq \sum_n (K/n^3 + 3K/n^2) < \infty$, hence $\sum_n S_n^4/n^4 < \infty$ a.s., forcing $\bar{X}_n \to 0$. ■

The weak law: for each large n, $\bar{X}_n$ is *probably* near μ; the strong law: a.s. the sequence eventually stays near μ. For $X_i = \mathbb{1}\{E \text{ on trial } i\}$, the long-run frequency of E is $P(E)$ a.s.

8.4 Other inequalities

Motivation. Chernoff bounds give exponential tails from the mgf; Jensen compares $E[g(X)]$ with $g(E[X])$.

Proposition 8.6 (One-sided Chebyshev)

If $E[X] = 0$ and $\mathrm{Var}(X) = \sigma^2$, then $P(X \geq a) \leq \sigma^2/(\sigma^2 + a^2)$ for $a > 0$.

Proof. For $b > 0$, $P(X \geq a) \leq P((X+b)^2 \geq (a+b)^2) \leq (\sigma^2 + b^2)/(a+b)^2$ by Markov; choose $b = \sigma^2/a$. ■

Proposition 8.7 (Chernoff bounds)

If $M(t) = E[e^{tX}]$, then $P(X \geq a) \leq e^{-ta}M(t)$ for all $t > 0$ and $P(X \leq a) \leq e^{-ta}M(t)$ for all $t < 0$.

Proof. For $t > 0$, $P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq E[e^{tX}]e^{-ta}$ by Markov. ■

Then minimize over t.

Example 8.4

(a) Standard normal: $e^{-ta+t^2/2}$ is minimized at $t = a$, so $P(Z \geq a) \leq e^{-a^2/2}$ for $a > 0$.
(b) Poisson(λ) and $i > \lambda$: minimizing $e^{\lambda(e^t-1)-ti}$ at $e^t = i/\lambda$ gives $P(X \geq i) \leq e^{-\lambda}(e\lambda)^i/i^i$.
(c) ± 1 fair plays: $M(t) = \cosh(t)^n \leq e^{nt^2/2}$, so $P(S_n \geq a) \leq e^{-a^2/(2n)}$ $(t = a/n)$; e.g. $P(S_{10} \geq 6) \leq 0.165$ (exact 0.0547).

Proposition 8.8 (Jensen's inequality)

If f is convex, then $E[f(X)] \geq f(E[X])$ whenever the expectations exist.

Proof. Convexity gives $f(x) \geq f(\mu) + f'(\mu)(x - \mu)$, $\mu = E[X]$; take expectations. ■

E.g. with concave utility u, $E[u(X)] \leq u(\mu)$: a sure return μ is preferred.

8.5 The Poisson approximation, quantified

For $X = \sum_{i=1}^n X_i$, independent $X_i \sim$ Bernoulli(p_i), and $Y \sim$ Poisson($\sum p_i$), coupling each X_i with a Poisson(p_i) variable agreeing w.p. $\geq 1 - p_i^2$ gives, for all A,

$$|P(X \in A) - P(Y \in A)| \leq \sum_{i=1}^n p_i^2.$$

For Bin(n, p) the bound is λp: good for small p, any n.

Example 8.5 (Birthdays, again)

With N the number of arrivals until the first shared birthday, $N > n$ iff no person $j \leq n$ shares with persons $1, \ldots, j-1$ (probability $(j-1)/365$ given no earlier match): $P(N > n) = \prod_{j=1}^n (1 - \frac{j-1}{365}) \approx e^{-n(n-1)/730}$, as in Example 4.6.

8.6 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): Weierstrass via the weak law, a bound for monotone pmfs, and the binomial Chernoff bound.

Exercise 8.4 (Convergence in probability and bounded functions)

Let $Z_1, Z_2, \dots$ be random variables and c a constant such that $P(|Z_n - c| > \epsilon) \to 0$ for every $\epsilon > 0$. Show that $E[g(Z_n)] \to g(c)$ for every bounded continuous g.

Solution.

Step 1. Let $|g| \le M$, $\epsilon > 0$, and $\delta > 0$ with $|g(x) - g(c)| < \epsilon$ for $|x - c| \le \delta$.

Step 2. Splitting on $\{|Z_n - c| \le \delta\}$,

$$|E[g(Z_n)] - g(c)| \le E|g(Z_n) - g(c)| \le \epsilon + 2M\, P(|Z_n - c| > \delta).$$

Step 3. So $\limsup_n |E[g(Z_n)] - g(c)| \le \epsilon$ for every ϵ. ■

Exercise 8.5 (Bernstein polynomials)

Let f be continuous on $[0, 1]$ and

$$B_n(x) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k} x^k (1-x)^{n-k}.$$

Prove that $B_n(x) \to f(x)$, uniformly in $x \in [0, 1]$. (This is a probabilistic proof of the Weierstrass approximation theorem.)

Solution.

Step 1. For $S_n \sim \mathrm{Bin}(n, x)$, $B_n(x) = E[f(S_n/n)]$, $E[S_n/n] = x$, $\mathrm{Var}(S_n/n) \le 1/(4n)$.

Step 2. $|f| \le M$ and f is uniformly continuous: $|f(u) - f(v)| < \epsilon$ for $|u - v| \le \delta$.

Step 3. As in Exercise 8.4 ($c = x$) with Chebyshev (Proposition 8.2),

$$|B_n(x) - f(x)| \le \epsilon + 2M\, P\Big(\Big|\frac{S_n}{n} - x\Big| > \delta\Big) \le \epsilon + \frac{2M}{4n\delta^2}.$$

Step 4. The bound is free of x, so $\limsup_n \sup_x |B_n(x) - f(x)| \le \epsilon$: uniform convergence, and polynomials are dense in $C[0, 1]$. ■

Exercise 8.6(a) (Monotone probability mass functions)

Let X take values in $\{1, 2, \dots\}$ with $P(X = k)$ nonincreasing in k. Prove that $P(X = k) \le 2E[X]/k^2$.

Solution.

Step 1. With $p_i = P(X = i)$,

$$E[X] = \sum_{i \ge 1} i p_i \ge \sum_{i=1}^{k} i p_i.$$

Step 2. $p_i \ge p_k$ for $i \le k$, so $\sum_{i=1}^{k} i p_i \ge p_k \frac{k(k+1)}{2} \ge p_k \frac{k^2}{2}$. ■

Exercise 8.11 (The Chernoff bound for the binomial)

Let $X \sim \mathrm{Bin}(n, p)$, $0 < p < 1$, $q = 1 - p$, and $np < i < n$.
(a) Show that $\min_{t>0} e^{-ti} E[e^{tX}]$ is attained where $e^t = iq/\{(n - i)p\}$.

(b) Deduce

$$P(X \geq i) \leq \frac{n^n}{i^i (n-i)^{n-i}}\, p^i q^{n-i}.$$

Solution.

(a)

Step 1. Minimize $h(t) = -ti + n\log(pe^t + q)$ over $t > 0$:

$$h'(t) = -i + \frac{npe^t}{pe^t + q}, \qquad h''(t) = \frac{npqe^t}{(pe^t+q)^2} > 0,$$

so h is strictly convex.

Step 2. $h'(t) = 0$ iff $e^t = iq/\{(n-i)p\}$, positive t since $i > np$.

(b) There $pe^t + q = \frac{nq}{n-i}$ and $e^{-ti} = \left(\frac{(n-i)p}{iq}\right)^i$; by Proposition 8.7

$$P(X \geq i) \leq \Big(\frac{(n-i)p}{iq}\Big)^i \Big(\frac{nq}{n-i}\Big)^n = \frac{n^n}{i^i(n-i)^{n-i}}\, p^i q^{n-i}.$$

E.g. $n = 100$, $p = \frac{1}{2}$, $i = 60$: bound 0.134, exact 0.028. ■